

Supplementary Material: CL-TAV Contracts, Wrappers and Decision Tables

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Abstract

This supplement is part of the delivered manuscript package. It typesets ALG-CLTAV-02/03/04 wrappers, ALG-CLTAV-05/06/07 cores, typed records and the A/P stop tables. It does not change ALG-CLTAV-01 and does not report experiments. Supplement algorithm numbers are S1, S2, ...; they are not the main-text Algorithm 1.

1 Scope and numbering

Main-text Algorithm 1 is ALG-CLTAV-01. Supplement Algorithms S1–S6 are ALG-CLTAV-02, 03, 04, 05, 06 and 07. S9 in the main text calls COMMITCOMPATIBLEUPDATE (S6 / ALG-CLTAV-07), not a bare IF-HIST-UPDATE invocation.

2 S0–S10 duty map

Step	Call	Duty / exception
S0	INITIALIZE	Γ, η, H, U ; distinct k , charges, retry
S1	IF-RESOURCE-STOP with IF-EQUIV	already decided Admit A2–A5 or P-stop
S2	PREDICTCURRENT	ALG-CLTAV-02; TEST-scoped GAP
S3	SELECTANDADMIT	ALG-CLTAV-05; freeze ξ including U
S4	CHARGEONCE	one charge; retry cap $\neq$ budget
S5	IF-EXECUTE-RECORD	raw record and effect class
S6	INTERPRETOUTCOME	ALG-CLTAV-03 + timing ALG-CLTAV-06
S7–S8	RESOLVEOUTCOME	ALG-CLTAV-04; adopt returned contexts
S9	COMMITCOMPATIBLEUPDATE	version/intersection; SPEC-ERROR; summary commit
S10	IF-RESOURCE-STOP with IF-EQUIV	P-stop or continue to S1

3 Typed records

Record	Fields
PredictionResult	status OK GAP; reason; classesByTest (OK); historyVersionUsed; uncertaintyRef
Decision	kind ACTION EXIT SPEC-ERROR; actionKind; actionId; classesUsed; reason
SelectSnapshot ξ	actionId; actionKind; qUsedAtSelect; historyVersion; classesUsedAtSelect; uncertaintyRef
ExecutionResult	attemptId; record; effect; correlationId
Outcome z	validity; effect; kind; actionId; I_z or NONE; prepResultEvaluated; targetConfirmed; summaryConfirmed; prepErr; declaredTarget; evidence; postSummary; timing
Resolution δ	control COMMIT RETRY STOP; Γ' ; η' ; outcome; reason

4 Admit and post-update stops

A1 selects one affordable, eligible, distinguishing action from the exclusive TEST, Prep or Recover alternatives. A2 is KNOWN with a distinguishing TEST or eligible Prep, but no such action affordable (Stop-Budget). A3 is KNOWN with no distinguishing TEST and no eligible Prep (Stop-NoDistinguisher). A4 is UNKNOWN with no eligible Recover (Stop-Error). A5 is UNKNOWN with Recover eligible but unaffordable (Stop-Budget). A TEST prediction GAP is handled independently: a KNOWN GAP without eligible Prep is SPEC-ERROR, not A3; retry-cap remains an independent Stop-Error condition. P1 Stop-Empty; P2 Stop-Singleton; P3 Stop-645; P4 Stop-Equivalent; P5 Stop-Budget after update. Stop-NoDistinguisher is exclusive with those P-stops when no remaining distinguisher exists and no P-stop already fired. Order is Empty, Singleton, 645, Equivalent, Budget. Singleton h_{normal} is not protocol PASS.

5 Short arguments

Monotonicity, retention (R1)–(R5), one-step minimax and resource termination are stated in the main text. The algebraic details are the accepted method §3.9.1 arguments: update is intersection; ERROR does not exclude; minimax is a least element of a finite selectable set; billed executions are bounded by the selected mode. They are not global optimality and not independent mathematical approval.

6 AI disclosure fields

System: PENDING-USER (AUTH-AI). Scope (text/figures/code): PENDING-USER. Author verification: required; not completed by this sentence. IEEE AECS guidance is rechecked at submission.

7 Wrappers and core mechanisms

Algorithm S1: PREDICTCURRENT: current-observation prediction wrapper (ALG-CLTAV-02, S2)

Require: SessionContext Γ ; HistoryHandle η ; current H ; uncertainty reference U ; action library L

Ensure: PredictionResult: status OK with nonempty classesByTest, or status GAP with a reason; the session’s own η is read, not rebuilt

```
1  $raw \leftarrow \text{IF-PRED-OBS}(\Gamma, \eta, H, U, L);$ 
  // branch on the backend tag BEFORE reading or projecting classes; a GAP result
  // carries no classesByTest
2 if  $raw.status = \text{GAP}$  then
3   | return PredictionResult(status = GAP, reason =  $raw.reason$ );
  // the gap stays TEST-scoped
4  $classes \leftarrow \text{PROJECTONTOCURRENTH}(raw.classes, H);$ 
  // a declared class set that projects to no nonempty class is the same named
  // gap, not OK and not  $score = 0$ 
5 if  $classes$  has no nonempty member then
6   | return PredictionResult(status = GAP, reason = no currently valid class);
7 return PredictionResult(status = OK, classesByTest =  $classes$ , historyVersionUsed =
   $\eta.version$ , uncertaintyRef =  $U$ );
```

Algorithm S2: INTERPRETOUTCOME: execution and observation coordination (ALG-CLTAV-03, S6)

Input: ExecutionResult er ; SelectSnapshot ξ ; Γ, η ; uncertainty reference U

Output: total Outcome z , including effect, validity, I_z , and confirmation fields

```

1  $z \leftarrow \text{NEWOUTCOME}()$ ;  $z.\text{effect} \leftarrow er.\text{effect}$ ;  $z.\text{kind} \leftarrow \xi.\text{actionKind}$ ;  $z.\text{actionId} \leftarrow \xi.\text{actionId}$ ;
2 if  $er.\text{effect} \in \{\text{CONFIRMED-NOT-SENT}, \text{UNKNOWN-EFFECT}\}$  then
3   return  $z$ ;
4  $(I_z, e, s, p, o, m) \leftarrow \text{IF-OBS-INTERPRET}(\Gamma, er.\text{record}, \text{clocks}, \varepsilon)$ ;
5 if  $o$  is invalid then
6   set  $z.\text{validity} \leftarrow \text{ERROR}$  and  $z.\text{effect} \leftarrow \text{UNKNOWN-EFFECT}$ ; return  $z$ ;
7  $\text{timing} \leftarrow \text{INTERPRETTIMEDOBSERVATION}(o, m, \xi, N_r, D_r, \text{Match}_r)$ ;
8 if  $\text{timing.verdict} = \text{ERROR}$  then
9   set  $z.\text{validity} \leftarrow \text{ERROR}$  and  $z.\text{effect} \leftarrow \text{UNKNOWN-EFFECT}$ ; return  $z$ ;
10 set  $z.\text{validity} \leftarrow \text{timing.verdict}$  and  $z.I_z \leftarrow I_z$ ;
11 if  $\xi.\text{actionKind} \in \{\text{PREP}, \text{RECOVER}\}$  then
12   // IF-PREP-RECOVER returns values only; it does not write  $\Gamma$  or  $\eta$ 
13    $z.\text{prepResultEvaluated} \leftarrow \text{true}$ ;
14    $(z.\text{targetConfirmed}, s, z.\text{prepErr}, z.\text{declaredTarget}, z.\text{evidence}, p) \leftarrow$ 
15    $\text{IF-PREP-RECOVER}(\Gamma, \eta, \xi.\text{actionId}, er.\text{record})$ ;
16 set  $z.\text{summaryConfirmed} \leftarrow s$  and  $z.\text{postSummary} \leftarrow p$ ;
17 if  $e = \text{UNKNOWN-EFFECT}$ ,  $z.\text{prepErr}$ , an unconfirmed Prep summary, an unconfirmed Recover
   target, or  $e = \text{NONE}$  with no confirmed summary then
18   set  $z.\text{effect} \leftarrow \text{UNKNOWN-EFFECT}$ ;
19 else
20   set  $z.\text{effect} \leftarrow e$ ;
21   // all  $\Gamma/\eta$  writes stay in ALG-CLTAV-04 and S9
22 return  $z$ ;
```

Algorithm S3: RESOLVEOUTCOME: ERROR, retry and confirmed recovery (ALG-CLTAV-04, S7–S8)

Require: Outcome z (effect, kind, validity, confirmation); SelectSnapshot ξ ; Γ ; η

Ensure: Resolution $\delta = (\text{control}, \Gamma', \eta', \text{outcome}, \text{reason})$: Γ' initialized from Γ

```

1  $\Gamma' \leftarrow \Gamma$ ;  $\eta' \leftarrow \eta$ ;
  // copy/persist semantics: the returned contexts are total, never undefined
2 if  $z.\text{validity} = \text{ERROR}$  or  $z.\text{effect} \in \{\text{CONFIRMED-NOT-SENT}, \text{UNKNOWN-EFFECT}\}$  or  $z.\text{prepErr}$  or
   ( $z.\text{kind}$  is Prep and  $z.\text{prepResultEvaluated}$  and  $z.\text{summaryConfirmed}$  is false) or ( $z.\text{kind}$  is
   Recover and  $z.\text{prepResultEvaluated}$  and  $z.\text{targetConfirmed}$  is false) then
  // S7 classify by kind and evaluated confirmation; do not fold
  CONFIRMED-NOT-SENT into UNKNOWN
3 do not exclude  $H$ ;
4 if  $z.\text{validity} = \text{ERROR}$  or  $z.\text{prepErr}$  or  $z.\text{effect} = \text{UNKNOWN-EFFECT}$  or ( $z.\text{kind}$  is Prep and
    $z.\text{prepResultEvaluated}$  and  $z.\text{summaryConfirmed}$  is false) then
5   | mark  $\eta'$  conservative-unknown;  $\Gamma'.\text{qStatus} \leftarrow \text{UNKNOWN}$ ; clear  $\Gamma'.\text{currentSummary}$ ;
6 else if  $z.\text{effect} = \text{CONFIRMED-NOT-SENT}$  then
7   | leave  $\Gamma'.\text{qStatus}$  and compatible history unchanged;
8 else
9   |  $\Gamma'.\text{qStatus}$  stays UNKNOWN; clear  $\Gamma'.\text{currentSummary}$ ; mark  $\eta'$  conservative-unknown;
   | // unconfirmed Recover target
10  $\Gamma'.\text{retryCount} \leftarrow \Gamma.\text{retryCount} + 1$ ;
11 if  $\Gamma'.\text{retryCount} \geq \Gamma'.\text{retryCap}$  then return
   Resolution( $\text{control} = \text{STOP}, \Gamma', \eta', z, \text{reason} = \text{Stop-Error}$ );
12 return Resolution( $\text{control} = \text{RETRY}, \Gamma', \eta', z$ );
   // an ERROR/unknown outcome reaches retry, never commit
13 else if  $z.\text{kind}$  is Recover then
  // S8 a confirmed Recover is a necessary condition for this branch; sole
  writer of KNOWN; the normalized summary returns to S9
14  $\Gamma'.\text{qStatus} \leftarrow \text{KNOWN}$ ;  $z.\text{postSummary} \leftarrow (z.\text{declaredTarget}, z.\text{evidence})$ ;
15 return Resolution( $\text{control} = \text{COMMIT}, \Gamma', \eta', z$ );
   // a normal TEST/Prep with a valid observation commits; the normalized Outcome
   carries the confirmed postSummary to S9

```

Algorithm S4: SELECTANDADMIT: test selection and resource admission (ALG-CLTAV-05, S3)

Require: Γ ; η ; current H ; action library L ; typed $pred$ (PredictionResult)

Ensure: a Decision: kind ACTION / EXIT / SPEC-ERROR with actionKind, actionId, classesUsed and reason

// state eligibility is independent of affordability; under UNKNOWN only an eligible Recover qualifies, never a Prep

```

1 eligible( $t$ )  $\leftarrow$  ( $\Gamma.qStatus = UNKNOWN ? (t.kind = RECOVER \wedge t.eligibleUnknown) : (t.kind \neq$ 
   RECOVER  $\wedge t.enabledAt = \Gamma.currentSummary)$ );
2  $A \leftarrow \{t \in L : eligible(t)\}$ ;
   // affordability is decided by IF-SELECT-ADMIT, not assumed from eligibility
   // a global or per-TEST projection gap withholds TEST scoring and keeps the gap
   identity; it does not itself block eligible Prep/Recover
3  $testGap \leftarrow false$ ;
4 if  $pred.status = GAP$  then
5    $C \leftarrow \{\}$ ;  $testGap \leftarrow true$ ;
6 else
7    $C \leftarrow \{C_t\}$  with  $C_t \leftarrow \{H \cap C : C \in pred.classesByTest[t], H \cap C \neq \emptyset\}$  for each TEST  $t \in A$ ;
8   if a TEST  $t \in A$  has declared classes but  $C_t = \emptyset$  then
9      $C \leftarrow \{\}$ ;
10     $testGap \leftarrow true$ ;
   // IF-SELECT-ADMIT is the single admission authority: affordability and A1-A5;
   exclusive kind; one-step minimax  $s(t) = \max_{C \in C_t} |C|$  over currently valid
   nonempty classes of affordable distinguishing TESTs  $D \cap F$  only; cost/id
   tie-break; never argmin  $\emptyset$ ; Prep/Recover are not observation minimax;
   uninformative TESTs are not A2
11 ( $k, S, t^*, admit$ )  $\leftarrow$  IF-SELECT-ADMIT( $A, \Gamma.currentSummary, C, \Gamma, \eta, H$ );
12 if  $k = ADMIT$  then
13   if  $testGap$  and  $admit = A3$  then return
       Decision(kind = SPEC-ERROR, reason =  $pred.reason$  or per-test prediction gap);
14   return Decision(kind = EXIT, reason = the A2-A5 admission reason);
15 else if  $k = PREP$  then
16   return Decision(kind = ACTION, actionKind = PREP, actionId =  $t^*$ , classesUsed =  $S$ );
17 else if  $k = RECOVER$  then
18   return Decision(kind = ACTION, actionKind = RECOVER, actionId =  $t^*$ ,  $\cdot$ );
19 else if  $testGap$  then
20   return Decision(kind = SPEC-ERROR, reason =  $pred.reason$ );
21 return Decision(kind = ACTION, actionKind = TEST, actionId =  $t^*$ , classesUsed =  $C_{t^*}$ );

```

Algorithm S5: INTERPRETTIMEDOBSERVATION: timed observation kernel (ALG-CLTAV-06, S6 kernel)

Require: the single ownership result and the raw measurement interval from
 IF-OBS-INTERPRET; SelectSnapshot ξ ; named obligation inputs N_r (allowed interval) and D_r (physical domain); declared match policy Match_r ; declared uncertainty ε

Ensure: a TimingResult: verdict in {PASS, FAIL, INCONCLUSIVE, ERROR}; the single pairing evidence; the interval used; validity

// resolve N_r , D_r and ε from the named obligation and instance inputs, not from free symbols

```

1 pair  $\leftarrow \text{Match}_r(\text{ownership}, \xi)$ ;
  // one pairing result feeds T2 and the no-response rule
2 if pair is ambiguous then return TimingResult(verdict = ERROR);
3 if pair is a legitimate trigger/response pair then
4    $M \leftarrow \text{measurementInterval} \cap D_r$ ;
5   if  $M = \emptyset$  then return TimingResult(verdict = ERROR);
6   if  $M \subseteq N_r$  then
7     return TimingResult(verdict = PASS, evidence = pair,  $M$ );
8   else if  $M \cap N_r = \emptyset$  then
9     return TimingResult(verdict = FAIL, evidence = pair,  $M$ );
10  return TimingResult(verdict = INCONCLUSIVE, evidence = pair,  $M$ );
    // open/closed endpoints preserved
  // a missing response forms a no-response only after a still-active instance
  // reaches its deadline; this round not seeing it is not a timeout
11 if the obligation is still active and expired then
    // use the earliest admissible elapsed horizon (a scalar), with the
    // requirement's boundary convention
12  if  $N_r$  is closed at  $U_r$  then
13    expired  $\leftarrow \text{lowerHorizon} > U_r$ ;
14  else
15    expired  $\leftarrow \text{lowerHorizon} \geq U_r$ ;
16  if expired then return TimingResult(verdict = FAIL);
17  return TimingResult(verdict = INCONCLUSIVE);
18 return TimingResult(verdict = ERROR);
    // a missing pair is not a point observation

```

Algorithm S6: COMMITCOMPATIBLEUPDATE: compatible update and summary commit (S6 / ALG-CLTAV-07)

Input: snapshot ξ ; Outcome z ; contexts Γ, η, H

Output: UpdateResult ($status, \eta', H', \Gamma'$)

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1  $\Gamma' \leftarrow \Gamma$ ;
2 if  $\xi.historyVersion \neq \eta.version$  then return SPEC-ERROR with unchanged  $\eta, H, \Gamma'$ ;
3 if  $z.I_z$  is a valid compatible class then
4    $(\eta_c, H_c) \leftarrow$  IF-HIST-UPDATE with;;
5    $\eta, H, \xi.actionId, \xi.qUsedAtSelect, \xi.classesUsedAtSelect,$ ;
6    $\xi.historyVersion, z.I_z, z.postSummary$ ;
7   if  $H_c \neq H \cap z.I_z$  then return SPEC-ERROR (intersection mismatch);
8    $\eta' \leftarrow \eta_c; H' \leftarrow H_c$ ;
9 else
10   $(\eta', H') \leftarrow$  IF-HIST-UPDATE with the same snapshot fields, NONE, and  $z.postSummary$ ;
11   $H' \leftarrow H$ ;
12 if  $\Gamma'.qStatus$  is KNOWN and  $z.summaryConfirmed$  then
13    $\Gamma'.currentSummary \leftarrow z.postSummary$ ;
13 return OK with  $(\eta', H', \Gamma')$ ;

```
